

Cal 2 - Exam Question 34

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Question 34

Evaluate

$$\int \frac{x^2}{(4-x^2)^{\frac{3}{2}}} dx.$$

- a. $\frac{2x}{\sqrt{4-x^2}} - 2 \sin^{-1}\left(\frac{x}{2}\right) + C$
- b. $\frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$
- c. $\frac{4}{\sqrt{4-x^2}} - 2 \sin^{-1}\left(\frac{x}{2}\right) + C$
- d. I do not know

Solution: b) $\frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$

This question is asking you to: Compute an [antiderivative](#) by rewriting the [denominator](#) into a form that matches a standard [trigonometric substitution](#) pattern, then substitute back to write the final answer in terms of x .

Step 1: Rewrite the denominator $(4-x^2)^{\frac{3}{2}}$.

Step 1.1:

Identify the [denominator](#)

In

$$\frac{x^2}{(4-x^2)^{\frac{3}{2}}},$$

the denominator is the bottom part:

$$(4-x^2)^{\frac{3}{2}}.$$

Step 1.2:Rewrite [fractional exponents](#)

Since the denominator is complicated, we want to simplify it by rewriting the power.

$$(4 - x^2)^{\frac{3}{2}}$$

has a fractional exponent, and we can rewrite fractional exponents using:

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m, \quad \sqrt[n]{a} = a^{\frac{1}{n}}.$$

Step 1.3:Rewrite the power $\frac{3}{2}$

$$\begin{aligned} \text{Rewrite } \frac{3}{2} &= \frac{1}{2} \cdot 3 & (4 - x^2)^{\frac{3}{2}} &= (4 - x^2)^{\frac{1}{2} \cdot 3} \\ \text{Use } a^{bc} &= (a^b)^c & (4 - x^2)^{\frac{1}{2} \cdot 3} &= \left((4 - x^2)^{\frac{1}{2}} \right)^3 \\ \text{Rewrite } (4 - x^2)^{\frac{1}{2}} &\text{ as a } \text{square root} & \left((4 - x^2)^{\frac{1}{2}} \right)^3 &= \left(\sqrt{4 - x^2} \right)^3 \end{aligned}$$

Step 2: Set up [trigonometric substitution](#).**Step 2.1:**

Notice the trig substitution pattern

Right now the denominator still looks complicated, so we want to simplify it more. The denominator contains

$$\sqrt{4 - x^2},$$

and this looks like the [trigonometric substitution](#) pattern

$$\sqrt{a^2 - x^2}.$$

Step 2.2:

Define the symbols

a : [constant](#), x : [variable](#), θ : angle.

Step 2.3:

Trigonometric pattern

$\sqrt{a^2 - x^2}$

When we encounter the pattern

$$\sqrt{a^2 - x^2},$$

we let

$$x = a \sin \theta.$$

Step 2.4:Rewrite the fraction $\sqrt{a^2 - x^2}$ Substitute $x = a \sin \theta$

$$\sqrt{a^2 - \boxed{x^2}} = \sqrt{a^2 - (\boxed{a \sin \theta})^2}$$

Step 2.5:Simplify $\sqrt{a^2 - (a \sin \theta)^2}$ Use [Power of a Product Rule](#), $(\boxed{a} \boxed{b})^2 = a^2 \boxed{b}^2$ Substitute $b = \sin \theta$, $(\boxed{a \sin \theta})^2 = a^2 (\boxed{\sin \theta})^2$

$$\sqrt{a^2 - \boxed{(a \sin \theta)^2}} = \sqrt{a^2 - \boxed{a^2 (\sin \theta)^2}}$$

Step 2.6:

Trig substitution summary

For pattern $\sqrt{a^2 - x^2}$, let $x = a \sin \theta$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 (\sin \theta)^2}$$

Step 3: Rewrite $\sqrt{4 - x^2}$ using trigonometric substitution.**Step 3.1:**Identify a and x in $\sqrt{4 - x^2}$ Rewrite $4 = 2 \cdot 2 = 2^2$

$$\sqrt{\boxed{4} - x^2} = \sqrt{\boxed{2^2} - x^2}$$

$$\text{Let } \sqrt{2^2 - x^2} = \sqrt{a^2 - x^2}$$

$$a = \boxed{2}$$

$$x = \boxed{a} \sin \theta = \boxed{2} \sin \theta$$

Step 3.2:Substitute $a = 2$, $x = 2 \sin \theta$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2(\sin \theta)^2}$$

into the fraction

$$\sqrt{2^2 - x^2} = \sqrt{2^2 - 2^2(\sin \theta)^2}$$

Step 4: Simplify $\sqrt{2^2 - 2^2(\sin \theta)^2}$.**Step 4.1:**Replace $2^2 = 4$

$$\text{Replace } 2^2 = 4 \quad \sqrt{2^2 - 2^2(\sin \theta)^2} = \sqrt{4 - 4(\sin \theta)^2}$$

Step 4.2:

Factor out 4

$$\text{Rewrite } 4 = 4 \cdot 1 \quad 4 - 4(\sin \theta)^2 = 4(1) - 4((\sin \theta)^2)$$

$$\text{Factor out 4} \quad 4 - 4(\sin \theta)^2 = 4(1 - (\sin \theta)^2)$$

Step 4.3:Simplify $1 - (\sin \theta)^2$

$$\text{By } (\sin \theta)^2 = \sin^2 \theta, \quad 4(1 - (\sin \theta)^2) = 4(1 - \sin^2 \theta)$$

$$\text{By [Pythagorean identity](#),} \quad 1 - \sin^2 \theta = \cos^2 \theta$$

$$\text{Substitute the Pythagorean identity,} \quad 4(1 - \sin^2 \theta) = 4(\cos^2 \theta)$$

Step 4.4:

Summary

$$\sqrt{4 - x^2} = \sqrt{2^2 - x^2} \quad \text{from Step 3.1}$$

$$= \sqrt{2^2 - 2^2(\sin \theta)^2} \quad \text{from Step 3.2}$$

$$= \sqrt{4(1 - \sin^2 \theta)} \quad \text{from Step 4.1 and Step 4.2}$$

$$= \sqrt{4 \cos^2 \theta} \quad \text{from Step 4.3}$$

Step 5: Compute $(4 - x^2)^{\frac{3}{2}}$ **after substitution.**

Step 5.1:**Substitute into** $(\sqrt{4-x^2})^3$

From Step 1.3, $(4-x^2)^{\frac{3}{2}} = (\sqrt{4-x^2})^3$

Substitute $\sqrt{4-x^2} = \sqrt{4\cos^2\theta}$ $\left(\sqrt{4-x^2}\right)^3 = \left(\sqrt{4\cos^2\theta}\right)^3$

Step 5.2:**Simplify** $\sqrt{4\cos^2\theta}$

Since $\sqrt{ab} = \sqrt{a}\sqrt{b}$, $\sqrt{4\cos^2\theta} = \sqrt{4}\sqrt{\cos^2\theta}$

$\sqrt{4} = 2$ $\sqrt{4}\sqrt{\cos^2\theta} = 2\sqrt{\cos^2\theta}$

Step 5.3:**Rewrite** $\sqrt{4\cos^2\theta}$ as $2|\cos\theta|$

Use $\sqrt{u^2} = |u|$ $\sqrt{\cos^2\theta} = |\cos\theta|$

Multiply both sides by 2 $2\sqrt{\cos^2\theta} = 2|\cos\theta|$

Step 5.4:**Cube both sides**

Cube both sides $\left(\sqrt{4\cos^2\theta}\right)^3 = (2|\cos\theta|)^3$

Use $(ab)^3 = a^3b^3$ $(2|\cos\theta|)^3 = 2^3|\cos\theta|^3$

Simplify 2^3 $\left(\sqrt{4\cos^2\theta}\right)^3 = 8|\cos\theta|^3$

Step 5.5:**Remove the absolute value**

For $x = 2\sin\theta$, we take $0 \leq \theta \leq \frac{\pi}{2}$ $\cos\theta \geq 0$

So $|\cos\theta| = \cos\theta$ $8|\cos\theta|^3 = 8\cos^3\theta$

Step 5.6:**Summary**

$$\sqrt{4 \cos^2 \theta} = 2\sqrt{\cos^2 \theta} \quad \text{from Step 5.2}$$

$$2\sqrt{\cos^2 \theta} = 2|\cos \theta| \quad \text{from Step 5.3}$$

$$\left(\sqrt{4 \cos^2 \theta}\right)^3 = 8|\cos \theta|^3 \quad \text{from Step 5.4}$$

$$(4 - x^2)^{\frac{3}{2}} = 8 \cos^3 \theta \quad \text{from Step 5.6}$$

Step 6: Rewrite the [numerator](#) x^2 .**Step 6.1:****Square both sides**

$$\text{Start with the substitution} \quad x = 2 \sin \theta \quad \text{from Step 3.1}$$

$$\text{Square both sides} \quad x^2 = (2 \sin \theta)^2$$

Step 6.2:**Apply $(ab)^2 = a^2 b^2$**

$$\text{Let } a = 2, b = \sin \theta \quad (ab)^2 = a^2 b^2$$

$$\text{Substitute into the rule} \quad (2 \sin \theta)^2 = 2^2 (\sin \theta)^2$$

Step 6.3:**Simplify $2^2 (\sin \theta)^2$**

$$\text{Compute } 2^2 \quad 2^2 = 4$$

$$\text{By } \text{square of trigonometric functions} \quad (\sin \theta)^2 = \sin^2 \theta$$

$$\text{Substitute} \quad 2^2 (\sin \theta)^2 = 4 \sin^2 \theta$$

$$\text{Therefore} \quad x^2 = 4 \sin^2 \theta$$

Step 7: Compute dx .

Step 7.1:**Differentiate** $x = 2 \sin \theta$ Start with the substitution $x = 2 \sin \theta$ from Step 3.1Differentiate both sides with respect to θ $\frac{dx}{d\theta} = \frac{d}{d\theta}(2 \sin \theta)$ Use the [constant multiple rule](#) $\frac{dx}{d\theta} = 2 \frac{d}{d\theta}(\sin \theta)$ Since $\frac{d}{d\theta}(\sin \theta) = \cos \theta$ $\frac{dx}{d\theta} = 2 \cos \theta$ **Step 7.2:****Write** dx Start with $\frac{dx}{d\theta} = 2 \cos \theta$ Multiply both sides by $d\theta$ $d\theta \left(\frac{dx}{d\theta} \right) = (2 \cos \theta) d\theta$ So $dx = 2 \cos \theta d\theta$ **Step 8: Rewrite the integral with substitutions.****Step 8.1:****List substitutions**

$$x^2 = 4 \sin^2 \theta \quad \text{from Step 6}$$

$$(4 - x^2)^{\frac{3}{2}} = 8 \cos^3 \theta \quad \text{from Step 1 to Step 5}$$

$$dx = 2 \cos \theta d\theta \quad \text{from Step 7}$$

Step 8.2:**Substitute**

$$\text{Start with the original integral} \quad \int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}} dx$$

$$\text{Substitute the expressions from Step 8.1} \quad \int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}} dx = \int \frac{4 \sin^2 \theta}{8 \cos^3 \theta} (2 \cos \theta d\theta)$$

Step 9: Simplify the integrand.

Step 9.1:

Multiply constants

$$\int \frac{4 \sin^2 \theta}{8 \cos^3 \theta} 2 \cos \theta \, d\theta = \int \frac{4 \cdot 2 \sin^2 \theta \cos \theta}{8 \cos^3 \theta} \, d\theta$$

Simplify $4 \cdot 2$

$$\int \frac{8 \sin^2 \theta \cos \theta}{8 \cos^3 \theta} \, d\theta = \int \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta$$

Step 9.2:

Cancel 8

Cancel the common factor 8

$$\int \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta = \int \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta$$

Step 9.3:Cancel $\cos \theta$

Rewrite $\cos^3 \theta$ $\cos^3 \theta = \cos^2 \theta \cos \theta$

Substitute into the denominator

$$\int \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta = \int \frac{\sin^2 \theta \cos \theta}{\cos^2 \theta \cos \theta} \, d\theta$$

Cancel $\cos \theta$

$$\int \frac{\sin^2 \theta \cos \theta}{\cos^2 \theta \cos \theta} \, d\theta = \int \frac{\sin^2 \theta}{\cos^2 \theta} \, d\theta$$

Step 9.4:Simplify $\int \frac{\sin^2 \theta}{\cos^2 \theta} \, d\theta$

By definition of tangent, $\tan \theta = \frac{\sin \theta}{\cos \theta}$

Square both sides $\tan^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta}$

Rewrite the integrand

$$\int \frac{\sin^2 \theta}{\cos^2 \theta} \, d\theta = \int \tan^2 \theta \, d\theta$$

Step 10: Integrate in θ .

Step 10.1:**Rewrite the integral**Use the [trigonometry identity](#) $\tan^2 \theta = \sec^2 \theta - 1$ Substitute into the integral $\int \tan^2 \theta \, d\theta = \int (\sec^2 \theta - 1) \, d\theta$ By [additive property of integrals](#) $\int (\sec^2 \theta - 1) \, d\theta = \int \sec^2 \theta \, d\theta - \int 1 \, d\theta$ **Step 10.2:****Integrate**Integrate $\sec^2 \theta$ $\int \sec^2 \theta \, d\theta = \tan \theta$ Integrate 1 $\int 1 \, d\theta = \theta$ Combine the result $\int \sec^2 \theta \, d\theta - \int 1 \, d\theta = \tan \theta - \theta + C$ Therefore $\int \tan^2 \theta \, d\theta = \tan \theta - \theta + C$ **Step 11: Convert $\tan \theta - \theta + C$ back into x .****Step 11.1:****Identify what to convert back**We now have the antiderivative in terms of θ , and we want to convert it back to x . To do that, we need to rewrite both $\tan \theta$ and θ in terms of x .**Step 11.2:****Find θ** Start with $x = 2 \sin \theta$ from Step 3.1Multiply both sides by $\frac{1}{2}$ $\frac{1}{2} x = \frac{1}{2} 2 \sin \theta$

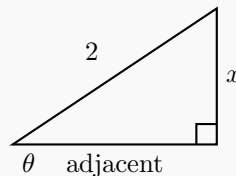
$$\frac{1}{2} \cdot 2 = 1 \quad \frac{x}{2} = \sin \theta$$

Apply [sin⁻¹](#) to both side of the equation $\theta = \sin^{-1}\left(\frac{x}{2}\right)$ **Step 11.3:****Write the sine ratio**Use [sin \$\theta = \frac{\text{opposite}}{\text{hypotenuse}}\$](#) $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{x}{2}$

Step 11.4:

Draw the triangle and identify the sides

From $\sin \theta = \frac{x}{2}$ opposite = x , hypotenuse = 2

**Step 11.5:**

Find the adjacent side

By the [Pythagorean Theorem](#), adjacent = $\sqrt{\text{hypotenuse}^2 - \text{opposite}^2}$

Substitute the values adjacent = $\sqrt{2^2 - x^2} = \sqrt{4 - x^2}$

Step 11.6:

Find $\tan \theta$

Use $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$ $\tan \theta = \frac{x}{\sqrt{4 - x^2}}$

Step 12: Write the final answer and choose the correct option.**Step 12.1:**

Substitute back

From Step 10.2, $\int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}} dx = \tan \theta - \theta + C$

Substitute Step 11.2 and Step 11.6 $\int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}} dx = \frac{x}{\sqrt{4 - x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$

Step 12.2:

Final answer

The correct option is **b**.

Definitions and Notation

Antiderivative

Definition: An antiderivative of a function is another function whose derivative gives back the original function. In simple words: if you take the derivative of the new function, you return to the old one.

Example:

$$\frac{d}{dx} \left(\frac{x^3}{3} \right) = x^2,$$

so one antiderivative of x^2 is $\frac{x^3}{3}$.

Denominator

Definition: In a fraction $\frac{p(x)}{q(x)}$, the denominator is the bottom part, $q(x)$.

Example:

$$\frac{x^2}{(4 - x^2)^{3/2}}$$

has numerator x^2 and denominator $(4 - x^2)^{3/2}$.

Numerator

Definition: In a fraction $\frac{p(x)}{q(x)}$, the numerator is the top part, $p(x)$.

Example:

$$\frac{x^2}{(4 - x^2)^{3/2}}$$

has numerator x^2 .

Constant

Definition: A constant is a fixed, unchanging value or number, such as 5, -10 , or π , that remains the same throughout a specific expression, equation, or function.

Example: In

$$3x + 7,$$

the 7 is the constant term.

Variable

Definition: A variable is a symbol, typically written in a lowercase letter like x , y , or z , that represents an unknown numerical value or a quantity that can change.

Example: In

$$3x + 7,$$

the x is the variable.

Power of a Product Rule

Definition: When raising a product of terms to a power, apply that power to each factor inside the parentheses, expressed as

$$(ab)^n = a^n b^n.$$

Example:

$$(3x)^2 = 3^2 x^2 = 9x^2.$$

Trigonometric Substitution

Definition: Trigonometric substitution is a method for simplifying integrals with square roots by replacing x with a trig expression involving an angle.

Example:

$$\sqrt{4 - x^2} = \sqrt{2^2 - x^2}$$

matches the pattern $\sqrt{a^2 - x^2}$, so we use $x = 2 \sin \theta$.

Common patterns:

$$\sqrt{a^2 - x^2} : x = a \sin \theta, \quad \sqrt{a^2 + x^2} : x = a \tan \theta, \quad \sqrt{x^2 - a^2} : x = a \sec \theta.$$

Fractional Exponents

Definition: Fractional exponents rewrite powers as roots.

$$a^{m/n} = \sqrt[n]{a^m} = \left(a^{1/n}\right)^m.$$

Example:

$$(4 - x^2)^{3/2} = \left((4 - x^2)^{1/2}\right)^3 = \left(\sqrt{4 - x^2}\right)^3.$$

This means the $3/2$ power tells us to take a square root first, then cube.

Square of Trigonometric Functions

Definition:

$$(\sin \theta)^2 = \sin^2 \theta, \quad (\cos \theta)^2 = \cos^2 \theta, \quad (\tan \theta)^2 = \tan^2 \theta.$$

Pythagorean Identity

Definition: The Pythagorean identity is the trig fact

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \Longleftrightarrow \quad 1 - \sin^2 \theta = \cos^2 \theta.$$

Example:

$$4 - 4 \sin^2 \theta = 4(1 - \sin^2 \theta) = 4 \cos^2 \theta.$$

Pythagorean Theorem

Definition: In a right triangle with legs a and b and hypotenuse c ,

$$a^2 + b^2 = c^2.$$

Example: If the hypotenuse is 2 and one side is x , then the other side satisfies

$$\text{adjacent}^2 = 2^2 - x^2 = 4 - x^2.$$

Square Root

Definition:

$$y^2 = x \implies y \text{ is a square root of } x.$$

For nonnegative real numbers,

$$\sqrt{x} = y \iff y^2 = x \text{ and } y \geq 0.$$

Example:

$$\sqrt{9} = 3$$

because

$$3^2 = 9.$$

Key Properties:

$$\sqrt{x^2} = |x|$$

$$\sqrt{a}\sqrt{b} = \sqrt{ab}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$(\sqrt{a})^2 = a \quad \text{and} \quad a^{\frac{1}{2}} = \sqrt{a}$$

Constant Multiple Rule

Definition: When taking a derivative, a constant factor can stay in front:

$$\frac{d}{d\theta}(c f(\theta)) = c f'(\theta).$$

Example:

$$\frac{d}{d\theta}(2 \sin \theta) = 2 \cos \theta.$$

Additive Property of Integrals

Definition: The integral of a sum or difference can be split into separate integrals:

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx$$

and

$$\int (f(x) - g(x)) dx = \int f(x) dx - \int g(x) dx.$$

Example:

$$\int (\sec^2 \theta - 1) d\theta = \int \sec^2 \theta d\theta - \int 1 d\theta.$$

Absolute Value

Definition: The absolute value of a number is its distance from 0, so it is never negative.

Example:

$$|3| = 3, \quad |-3| = 3.$$

In Q34 we use

$$\sqrt{u^2} = |u|.$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we know $\cos \theta \geq 0$, so

$$|\cos \theta| = \cos \theta.$$

Basic Trig Ratios: Sine, Cosine, and Tangent

Definition: In a right triangle:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Example: If $\sin \theta = \frac{x}{2}$, then opposite = x and hypotenuse = 2. The adjacent side is

$$\sqrt{2^2 - x^2} = \sqrt{4 - x^2},$$

so

$$\tan \theta = \frac{x}{\sqrt{4 - x^2}}.$$

Inverse Sine $\sin^{-1}(x)$

Definition: Also written $\arcsin(x)$, inverse sine returns the angle whose sine is x .

Example: If $\sin \theta = \frac{x}{2}$, then

$$\theta = \sin^{-1}\left(\frac{x}{2}\right).$$